

Timoteo Costantini Scala



Software Developer

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github.com/teemoteeo · teemoteeo.github.io/timoteo-site

Software engineering student at 42 Firenze with every Common Core project completed through Milestone 3, working in C and Python on concurrency, algorithms and LLM tooling. Also builds websites for clients as a freelance developer. Years in fashion and fine-dining hospitality have made me comfortable working under pressure and with demanding clients.

TECHNICAL EXPERIENCE

42 Firenze — Software Engineering, Common Core

Nov 2025 - Present

Project-based, peer-reviewed curriculum. Every project completed through Milestone 3.

Call Me Maybe Python · LLM · Qwen3-0.6B

- Function-calling pipeline based on constrained decoding. DFA-based token masking guarantees the output is always schema-valid.
- Detects repetition loops and uses logits to pick between function names that share a prefix.

Codexion C · pthreads

- Multithreaded resource scheduler (a dining-philosophers variant) with FIFO and EDF priority queues for each resource.
- Prevents deadlock through ordered lock acquisition. Runs clean under ThreadSanitizer.

A-Maze-ing Python · curses

- Maze generator and solver (DFS, Prim, Kruskal, BFS) with a terminal UI, packaged as a pip module. Pair project.

Fly-In Python · Pydantic · pytest · mypy

- Turn-based simulation that routes a fleet of drones through a weighted zone graph in the fewest turns. Zone and link capacities are enforced every turn.
- Uses weighted Dijkstra and k-distinct-path search with no graph libraries, plus deadlock detection. Parser for a custom map format with line-level error reporting.

Also: C fundamentals (memory management, Unix, algorithms) · Python OOP and design patterns (Abstract Factory, Strategy, Mixins).

Web Development — Freelance and personal projects

2026

HDGR Architecture hdgr.eu · HTML/CSS/JS · GSAP · Vercel

- Designed and built an architecture studio's website with scroll-driven GSAP animations. Self-hosted all assets so it loads reliably in mainland China.

Personal portfolio teemoteeo.github.io/timoteo-site · Vanilla JS · Three.js

- Terminal-style site with a boot sequence, typing animations and a Three.js Rubik's cube hidden behind the Konami code. No framework.

SKILLS

Languages C, Python, JavaScript, HTML/CSS

Concepts Concurrency and synchronization, algorithms and data structures, OOP and design patterns

Tools Git, Linux / command line, Docker, SQL

LLM tooling Constrained decoding, local inference (MLX), MCP

Spoken Italian (native), English (C2), German (basic)

EDUCATION

Università degli Studi di Milano — Computer Science for Digital Communication (2 years, no degree)

OTHER EXPERIENCE

Fashion Model — Self-employed · Valentino, Zegna, Prada, Oakley, Slam Jam · Paris Fashion Week

2022 - Present

Commis de salle — Giorgio Armani Michelin-starred restaurant, Paris (full-time)

Jan - Oct 2025

Waiter — Tram Tram, Milan (on-call)

2021 - 2024